The cloglog GLMM, step by step

A plain-language companion to the Technical Background section on the generalised linear mixed model. The numbers are made up for illustration and are computed exactly.

0. What the general words mean in this thesis

Text says	In this thesis
unit *j*	one coronary segment, e.g. the proximal LAD
cluster *i*	one person, with up to nine segments
event	new plaque in that segment by the repeat scan
window	the time from the baseline scan to the repeat scan
covariates *x*	curvature, taper, bifurcation angle, daughter asymmetry, SCORE2 risk factors

1. From a risk score to a probability

The model turns the features of a segment into a probability in three steps:

$$\eta_{ij} = \beta_0 + \sum_k \beta_k x_{ijk} \quad \longrightarrow \quad \Lambda_{ij} = e^{\eta_{ij}} \quad \longrightarrow \quad p_{ij} = 1 - e^{-\Lambda_{ij}}$$

Step 1, the risk score η. An ordinary regression sum: a starting value β_0 plus each feature times its weight β_k. It can be any number, positive or negative.

Step 2, the accumulated hazard $\Lambda = e^{\eta}$. Taking the exponential makes the score positive. It is the amount of risk built up over the window, roughly the expected number of plaques the segment would get.

Step 3, the probability $p = 1 - e^{-\Lambda}$. This turns the amount of risk into the chance of at least one event. When Λ is small, p is almost equal to Λ.

	Score η	Hazard $\Lambda = e^{\eta}$	Probability p
Segment A	–3.0	0.050	4.9%
Segment A, one unit more curved ($\beta = 0.4$)	–2.6	0.074	7.2%

Going backwards gives the name of the model: from p, take $\log(-\log(1 - p))$ and you are back at η. That function is the *complementary log-log* (cloglog) link.

2. Why $\exp(\beta_k)$ is a hazard ratio

Raise covariate k by one unit and keep everything else fixed. The score gains β_k, and because the exponential turns a sum into a product, the hazard is multiplied by a fixed factor:

$$e^{\eta + \beta_k} = e^{\eta} \cdot e^{\beta_k} \qquad \Rightarrow \qquad \frac{\Lambda_{\text{new}}}{\Lambda_{\text{old}}} = e^{\beta_k}$$

In the example, Λ went from 0.050 to 0.074, a factor of 1.49, and $e^{0.4} = 1.49$. The factor is the same whatever the starting score, which is why one number, $\exp(\beta_k)$, describes the effect of covariate k for every segment.

3. Why a hazard ratio and not an odds ratio

Follow-up differs between participants. A hazard ratio stays the same however long the window is; an odds ratio does not. Take a feature that truly doubles the hazard, with a baseline risk of 5% per year:

Window	Risk without feature	Risk with feature	Hazard ratio	Odds ratio
1 year	5.0%	9.8%	2	2.05
10 years	40.1%	64.2%	2	2.67

The biology is identical, but the odds ratio drifts from 2.05 to 2.67 only because the window is longer. A logit model pooling short and long follow-up would estimate a blend of the two. With the cloglog model, β_k means the same for everyone. When events are rare, as in the 1-year row, the two ratios almost agree, which is why the two links give nearly the same fit then.

4. Exposure: segments of different length

A longer segment has more wall on which plaque can form. Its length E enters the score as log E with its own weight γ:

$$\eta_{ij} = \beta_0 + \sum_k \beta_k x_{ijk} + \gamma \log E_{ij} \qquad \Rightarrow \qquad \Lambda_{ij} \propto E_{ij}^{\gamma}$$

With $\gamma = 1$ (an *offset*, fixed in advance), twice the length gives twice the hazard. With γ estimated, the data decide how strongly length matters.

5. The random effect: one hidden value per person

Segments of the same person share things that are not measured, such as genes, diet and lipid levels over the years. If one of their segments develops plaque, the others are more likely to as well. The model gives each person a hidden term b_i, added to the score of *all* their segments:

$$p_{ij}(b_i) = 1 - \exp\{-e^{\eta_{ij} + b_i}\}, \qquad b_i \sim \mathcal{N}(0, \sigma^2)$$

Person's b_i	Effect on every segment of that person
+0.7	hazard × 2.01
0	hazard unchanged (a typical person)
–0.7	hazard × 0.50

The individual b_i are not estimated. The model estimates σ^2, how much people differ. $\sigma^2 = 0$ means that people do not differ beyond the covariates.

> **Why it matters.** Without b_i, nine segments of one person count as nine independent pieces of evidence. They are not, so the model would be over-confident: standard errors too small and tests rejecting too often.

Because the coefficients are read with b_i held fixed, $\exp(\beta_k)$ compares two segments of the *same* person. Averaged over people, the effect looks smaller. σ^2 is hard to read on its own, so it is often translated into a median hazard ratio: the typical factor between the hazard of an identical segment in a higher-risk and a lower-risk person.

6. The nested version: arteries within a person

Segments of the same artery may be more alike than segments of different arteries. A second hidden term c, one per artery, captures this:

$$p_{ij} = 1 - \exp\{-e^{\eta_{ij} + b_i + c_{i\,g(j)}}\}, \qquad c_{ig} \sim \mathcal{N}(0, \omega^2)$$

Two segments of the same artery share both b and c; segments of different arteries share only b. $\omega^2 = 0$ gives back the simpler model. ω^2 is even harder to estimate than σ^2, as only arteries with more than one event inform it.

7. Fitting the model: quadrature

b_i is never observed, so the model averages over every value it could take. That average is an integral with no exact formula. Adaptive Gauss–Hermite quadrature replaces it with a weighted sum over a few points (nodes) placed around the most likely value. One node is the Laplace approximation, which is biased for small clusters; nodes are added until the estimates stop changing.

8. The main test: does geometry add anything?

The model is fitted twice on the same rows with the same settings: once without the geometry block and once with it. The log-likelihood *l* measures how well each fits:

$$\mathrm{LR} = 2\,(\ell_1 - \ell_0) \overset{H_0}{\sim} \chi^2_q$$

If geometry adds nothing, LR behaves like a χ^2 value with q degrees of freedom, q being the number of geometry terms (**q = 4** after Finet's ratio was dropped). A large LR means geometry improves the fit. The χ^2 reference only holds if the random effects capture the clustering; otherwise the test rejects too often.

9. From segments to a person

The clinical question is often whether a *person* develops any new plaque. Given b_i, the person stays plaque-free only if every segment does, so the probabilities multiply. With the cloglog link the product collapses into one exponential:

$$P(\text{none} \mid b_i) = \prod_{j=1}^{n_i} e^{-e^{\eta_{ij} + b_i}} = \exp\left(-e^{b_i} \sum_j e^{\eta_{ij}}\right) = e^{-e^{b_i} H_i}$$

> **The segment hazards simply add up:** $H_i = \Lambda_{i1} + \Lambda_{i2} + \dots$ This works because $e^{-a} \cdot e^{-b} = e^{-(a+b)}$. Odds do not add like this, which is a practical reason for the cloglog link.

Example. Nine segments with $\Lambda = 0.05$ each give $H = 0.45$. With $b = 0$, $P(\text{no plaque}) = e^{-0.45} = 0.64$, so the person's risk is 36.2%.

For a new person b_i is unknown, so the risk is averaged over all possible values of b:

$$P_i = 1 - \int e^{-e^{b} H_i} f(b; \sigma^2)\, \mathrm{d}b$$

With $\sigma^2 = 1$ and the same H, the averaged risk is 41.2%, against 36.2% when b is simply set to 0. Setting b = 0 gives the risk of the *median* person, not the average one. It ranks people in the same order, so only a calibration check reveals the difference. Multiplying segment risks as if they were independent is the other tempting shortcut, and it overstates the risk when plaque clusters within people.

10. Dilution

A feature that acts on one segment changes only that segment's share of H. Doubling the hazard of one of the nine segments (0.05 to 0.10) moves H from 0.45 to 0.50, and the person's risk only from 36.2% to 39.3%.

> **Reading a significant LR test.** Geometry can improve the fit at segment level, by predicting *which* segments get plaque, while barely changing a person's overall risk. A rejected test means the first; it does not by itself mean the second.

The whole model in one line

Step	What happens
Features	$\eta = \beta_0 + \Sigma\, \beta_k x_k$ (+ γ log length)
Person	add b_i (and c for the artery)
Hazard	$\Lambda = e^{\eta + b}$; one unit more of x_k multiplies it by e^{β_k}
Segment risk	$p = 1 - e^{-\Lambda}$
Person risk	add the Λ of all segments, then $1 - e^{-H}$, averaged over b
Test	$LR = 2(l_1 - l_0)$ against χ^2_4